Moderation Analysis: Schooling × Fossil-fuel share of Electricity

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1 Packages

```r
library(dplyr)
library(readr)
library(tidyr)
library(ggplot2)
library(car)
library(sandwich)
library(lmtest)
library(stargazer)
library(DescTools)
library(interactions)
library(knitr)
library(here)
```

2 Paths

```r
path_energy <- here("owid-energy-data.csv")
path_co2    <- here("owid-co2-data.csv")
path_school <- here("average_years_of_schooling.csv")
path_gdp    <- here("GDP_per_capita.csv")

target_year <- 2023
```

```r
energy <- read_csv(path_energy)
co2    <- read_csv(path_co2)
school <- read_csv(path_school)
gdp    <- read_csv(path_gdp)
```

3 Data cleaning

3.1 Fossilfuel share data

```r
fossil_share_elec <- read_csv(path_energy, show_col_types = FALSE)

fossil_share_elec_2023 <- fossil_share_elec %>%
filter(nchar(iso_code) == 3, year == target_year) %>%
select(country, iso_code, year, fossil_share_elec)

head(fossil_share_elec_2023)
```

country	iso_code	year	fossil_share_elec
<chr>	<chr>	<dbl>	<dbl>
Afghanistan	AFG	2023	13.402
Albania	ALB	2023	0.000
Algeria	DZA	2023	99.055
American Samoa	ASM	2023	100.000
Angola	AGO	2023	23.579
Antarctica	ATA	2023	NA

6 rows

```r
write_csv(
fossil_share_elec_2023,
here("output", paste0("owid-fossil-data-cleaned-", target_year, ".csv"))
)
```

3.2 CO2 per capita data

```r
co2pc <- read_csv(path_co2, show_col_types = FALSE)

co2_per_capita_2023 <- co2pc %>%
filter(nchar(iso_code) == 3, year == target_year) %>%
select(country, iso_code, year, co2_per_capita)

head(co2_per_capita_2023)
```

country	iso_code	year	co2_per_capita
<chr>	<chr>	<dbl>	<dbl>
Afghanistan	AFG	2023	0.254
Albania	ALB	2023	1.571
Algeria	DZA	2023	4.394
Andorra	AND	2023	5.170
Angola	AGO	2023	0.590
Anguilla	AIA	2023	9.879

6 rows

```r
write_csv(
co2_per_capita_2023,
here("output", paste0("owid-co2pc-data-cleaned-", target_year, ".csv"))
)
```

3.3 Mean years of schooling data

```r
schooling_2023 <- read_csv(path_school, show_col_types = FALSE) %>%
transmute(
country   = Entity,
iso_code  = iso_code,
year      = year,
schooling = as.numeric(schooling)
) %>%
filter(nchar(iso_code) == 3, year == target_year)

head(schooling_2023)
```

country	iso_code	year	schooling
<chr>	<chr>	<dbl>	<dbl>
Afghanistan	AFG	2023	2.514790
Albania	ALB	2023	10.175311
Algeria	DZA	2023	7.416387
Andorra	AND	2023	11.610000
Angola	AGO	2023	5.956173
Antigua and Barbuda	ATG	2023	11.647500

6 rows

```r
write_csv(
schooling_2023,
here("output", paste0("schooling-cleaned-", target_year, ".csv"))
)
```

3.4 GDP per capita data

```r
lines <- readLines(path_gdp, n = 300, warn = FALSE)
header_line <- which(grepl("Country Name", lines, fixed = TRUE))[1]

gdp_raw <- read_csv(path_gdp, skip = header_line - 1, show_col_types = FALSE)

year_idx <- match(as.character(target_year), names(gdp_raw))
if (is.na(year_idx)) stop("Year column not found. Please check if this file contains the target ye
ar.")

gdp_country_2023 <- tibble(
country = gdp_raw[[1]],
iso_code = gdp_raw[[2]],
year = target_year,
gdp_per_capita = as.numeric(gdp_raw[[year_idx]])
) %>%
filter(nchar(iso_code) == 3)

head(gdp_country_2023)
```

country	iso_code	year	gdp_per_capita
<chr>	<chr>	<dbl>	<dbl>
Aruba	ABW	2023	31178.4737
Africa Eastern and Southern	AFE	2023	1412.6254
Afghanistan	AFG	2023	378.0663
Africa Western and Central	AFW	2023	1841.7583
Angola	AGO	2023	2334.6610
Albania	ALB	2023	5444.9300

6 rows

```r
write_csv(
gdp_country_2023,
here("output", paste0("gdp-per-capita-cleaned-", target_year, ".csv"))
)
```

4 Data merging

```r
merged_yr <- fossil_share_elec_2023 %>%
left_join(
co2_per_capita_2023 %>% select(iso_code, year, co2_per_capita),
by = c("iso_code", "year")
) %>%
left_join(
schooling_2023 %>% select(iso_code, year, schooling),
by = c("iso_code", "year")
) %>%
left_join(
gdp_country_2023 %>% select(iso_code, year, gdp_per_capita),
by = c("iso_code", "year")
) %>%
drop_na(fossil_share_elec, co2_per_capita, schooling, gdp_per_capita)
head(merged_yr)
```

country	iso_code	year	fossil_share_elec	co2_per_capita	schooling ▸
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
Afghanistan	AFG	2023	13.402	0.254	2.514790
Albania	ALB	2023	0.000	1.571	10.175311
Algeria	DZA	2023	99.055	4.394	7.416387
Angola	AGO	2023	23.579	0.590	5.956173
Antigua and Barbuda	ATG	2023	94.444	6.863	11.647500
Argentina	ARG	2023	59.783	3.918	11.184582

6 rows | 1-6 of 7 columns

```r
write_csv(
merged_yr,
here("output", paste0("merged_dataset_", target_year, "_complete_cases.csv"))
)
```

5 Descriptive statistics

```r
df_desc <- merged_yr %>%
select(fossil_share_elec, co2_per_capita, schooling, gdp_per_capita) %>%
na.omit()

cat("Number of observations:", nrow(df_desc), "\n\n")
```

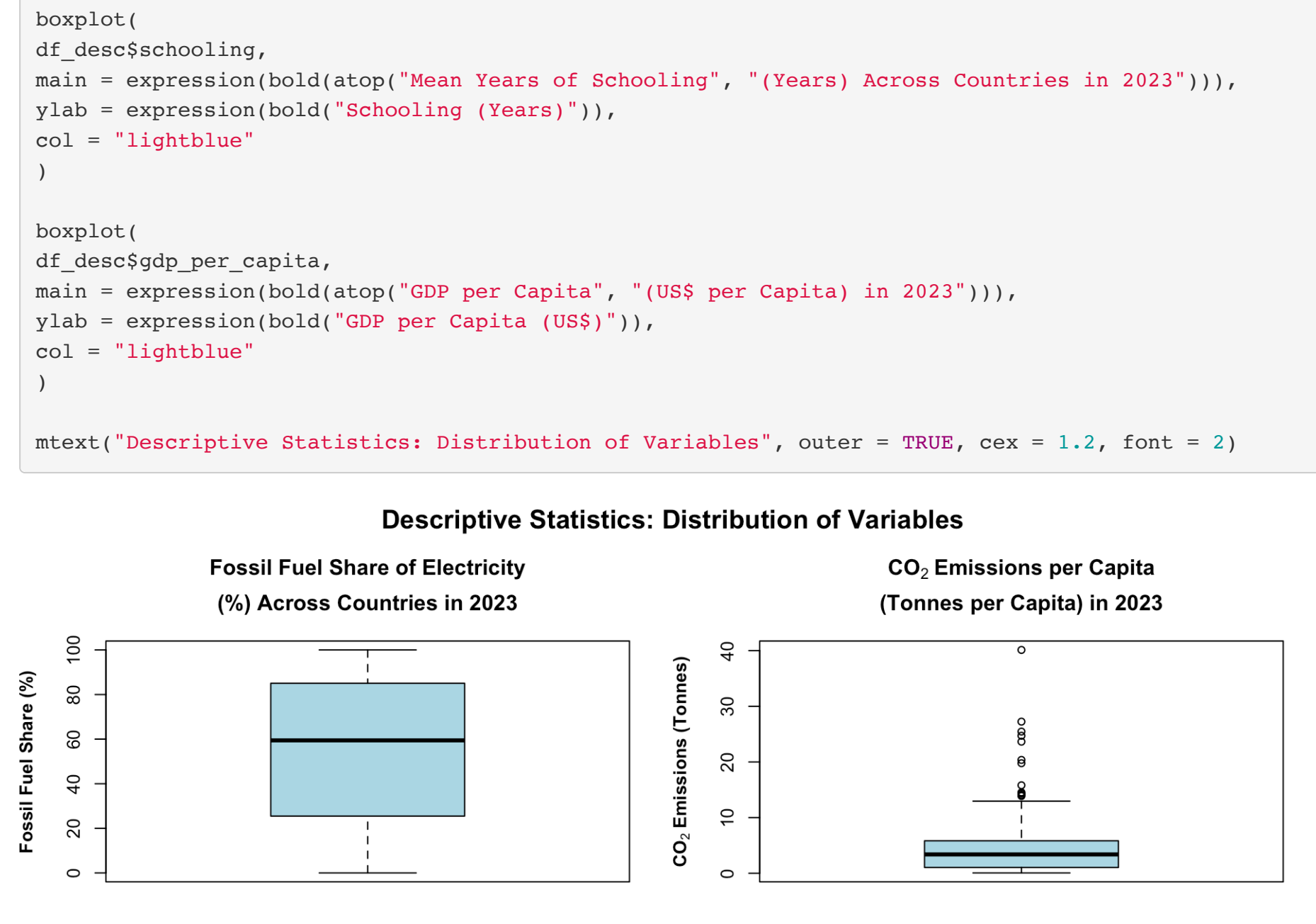
```
## Number of observations: 160
```

```r
summary(df_desc)
```

```
##  fossil_share_elec co2_per_capita     schooling      gdp_per_capita
##  Min.   :  0.00    Min.   : 0.056   Min.   : 1.412   Min.   :   253.4
##  1st Qu.: 25.50    1st Qu.: 1.060   1st Qu.: 7.359   1st Qu.:  2490.6
##  Median : 59.44    Median : 3.381   Median :10.131   Median :  6495.5
##  Mean   : 55.10    Mean   : 4.710   Mean   : 9.412   Mean   : 15441.0
##  3rd Qu.: 84.71    3rd Qu.: 5.753   3rd Qu.:11.845   3rd Qu.: 18859.5
##  Max.   :100.00    Max.   :40.128   Max.   :14.296   Max.   :104590.1
```

5.1 Boxplots

```r
op <- par(no.readonly = TRUE)
par(mfrow = c(2, 2), mar = c(4, 4, 5, 2), oma = c(0, 0, 2, 0))

boxplot(
df_desc$fossil_share_elec,
main = expression(bold(atop("Fossil Fuel Share of Electricity", "(%) Across Countries in 2023"))),
ylab = expression(bold("Fossil Fuel Share (%)")),
col = "lightblue"
)

boxplot(
df_desc$co2_per_capita,
main = expression(bold(atop(CO[2]~"Emissions per Capita", "(Tonnes per Capita) in 2023"))),
ylab = expression(bold(CO[2]~"Emissions (Tonnes)")),
col = "lightblue"
)

boxplot(
df_desc$schooling,
main = expression(bold(atop("Mean Years of Schooling", "(Years) Across Countries in 2023"))),
ylab = expression(bold("Schooling (Years)")),
col = "lightblue"
)

boxplot(
df_desc$gdp_per_capita,
main = expression(bold(atop("GDP per Capita", "(US$ per Capita) in 2023"))),
ylab = expression(bold("GDP per Capita (US$)")),
col = "lightblue"
)

mtext("Descriptive Statistics: Distribution of Variables", outer = TRUE, cex = 1.2, font = 2)
```



```r
par(op)
```

5.2 Correlation matrix

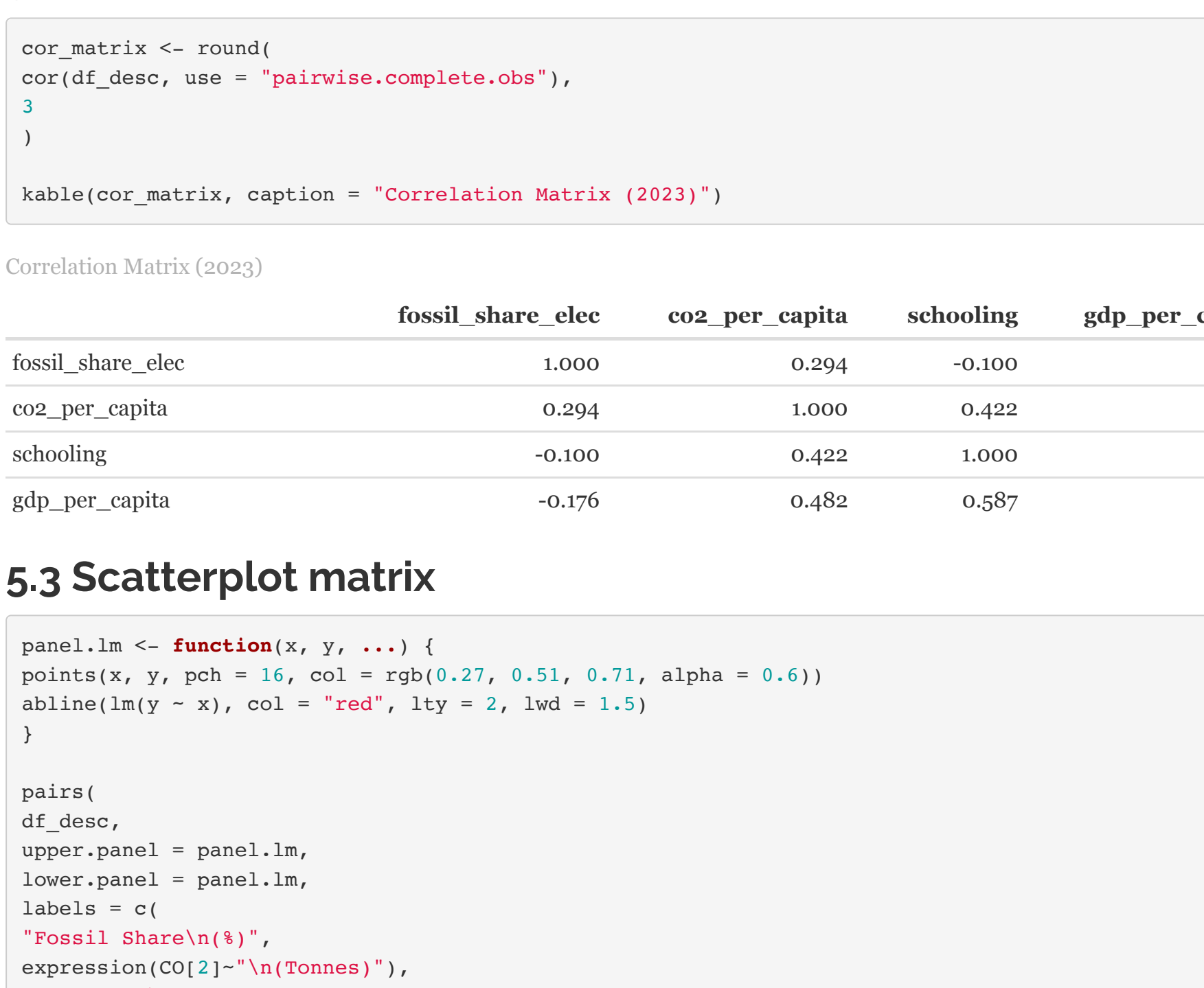
```r
cor_matrix <- round(
cor(df_desc, use = "pairwise.complete.obs"),
3
)

kable(cor_matrix, caption = "Correlation Matrix (2023)")
```

Correlation Matrix (2023)

	fossil_share_elec	co2_per_capita	schooling	gdp_per_capita
fossil_share_elec	1.000	0.294	-0.100	-0.176
co2_per_capita	0.294	1.000	0.422	0.482
schooling	-0.100	0.422	1.000	0.587
gdp_per_capita	-0.176	0.482	0.587	1.000

5.3 Scatterplot matrix

```r
panel.lm <- function(x, y, ...) {
points(x, y, pch = 16, col = rgb(0.27, 0.51, 0.71, alpha = 0.6))
abline(lm(y ~ x), col = "red", lty = 2, lwd = 1.5)
}

pairs(
df_desc,
upper.panel = panel.lm,
lower.panel = panel.lm,
labels = c(
"Fossil Share\n(%)",
expression(CO[2]~"\n(Tonnes)"),
"Schooling\n(Years)",
"GDP per Cap\n(US$)"
),
main = "Scatterplot Matrix with Linear Trend Lines"
)
```



6 Modelling

6.1 Winsor function and model dataframe

```r
winsor <- function(x, probs = c(0.01, 0.99)) {
qs <- quantile(x, probs = probs, na.rm = TRUE, type = 7)
pmin(pmax(x, qs[1]), qs[2])
}

df_model <- merged_yr %>%
transmute(
fossil_share = as.numeric(fossil_share_elec),
co2_pc       = as.numeric(co2_per_capita),
schooling    = as.numeric(schooling),
gdp_pc       = as.numeric(gdp_per_capita)
) %>%
filter(complete.cases(fossil_share, co2_pc, schooling, gdp_pc)) %>%
mutate(
co2_pc_w = winsor(co2_pc, probs = c(0.01, 0.99)),
co2_shift = ifelse(min(co2_pc_w, na.rm = TRUE) <= 0,
abs(min(co2_pc_w, na.rm = TRUE)) + 1e-6, 0),
ln_co2_pc = log(co2_pc_w + co2_shift),
ln_gdp = log(gdp_pc),
fossil_share_c = fossil_share - mean(fossil_share, na.rm = TRUE),
schooling_c    = schooling    - mean(schooling, na.rm = TRUE)
)
```

6.2 Three models

```r
m1 <- lm(ln_co2_pc ~ fossil_share_c + ln_gdp, data = df_model)
m2 <- lm(ln_co2_pc ~ fossil_share_c + ln_gdp + schooling_c, data = df_model)
m3 <- lm(ln_co2_pc ~ fossil_share_c + ln_gdp + schooling_c + fossil_share_c:schooling_c,data = df_
model)
```

6.3 Robust standard errors (HC1)

```r
se_m1_robust <- sqrt(diag(vcovHC(m1, type = "HC1")))[names(coef(m1))]
se_m2_robust <- sqrt(diag(vcovHC(m2, type = "HC1")))[names(coef(m2))]
se_m3_robust <- sqrt(diag(vcovHC(m3, type = "HC1")))[names(coef(m3))]
```

6.4 Regression table

```r
stargazer(
  m1, m2, m3,
  type = "html",
  title = "<span style='color: black; font-weight: bold;'> Table 2. Regression results of three mo
dels",
  dep.var.labels = "log(CO2 emissions per capita)",
  se = list(se_m1_robust, se_m2_robust, se_m3_robust),
  digits = 3,
  model.numbers = TRUE,
  order = c("fossil_share_c", "ln_gdp", "schooling_c", "fossil_share_c:schooling_c"),
  covariate.labels = c(
    "Fossil fuel share of electricity (centered)",
    "log(GDP per capita)",
    "Average years of schooling (centered)",
    "Fossil share × Schooling"
  ),
  notes.append = FALSE,
  notes = "Significance levels: * p < 0.05; ** p < 0.01; *** p < 0.001. <br> Robust standard error
s (HC1) in parentheses.",
  header = FALSE
)
```

Table 2. Regression results of three models

	Dependent variable:		
	log(CO2 emissions per capita)		
	(1)	(2)	(3)
Fossil fuel share of electricity (centered)	0.010^{***}	0.010^{***}	0.011^{***}
	(0.002)	(0.002)	(0.002)
log(GDP per capita)			0.001^{**}
			(0.0005)
Average years of schooling (centered)	0.839^{***}	0.615^{***}	0.633^{***}
	(0.041)	(0.067)	(0.071)
Fossil share × Schooling		0.120^{***}	0.118^{***}
		(0.028)	(0.028)
Constant	-6.558^{***}	-4.576^{***}	-4.723^{***}
	(0.373)	(0.598)	(0.627)
Observations	160	160	160
R^2	0.775	0.802	0.811
Adjusted R^2	0.772	0.799	0.806
Residual Std. Error	0.633 (df = 157)	0.595 (df = 156)	0.584 (df = 155)
F Statistic	270.607^{***} (df = 2; 157)	211.239^{***} (df = 3; 156)	166.210^{***} (df = 4; 155)
Note:	Significance levels: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.<br>Robust standard errors (HC1) in parentheses.		

6.5 Simple slope analysis

```r
s_slopes <- sim_slopes(
  model = m3,
  pred  = "fossil_share_c",
  modx  = "schooling_c",
  johnson_neyman = TRUE,
  cond.int = TRUE,
  robust = TRUE
)
print(s_slopes)
```

```
## JOHNSON-NEYMAN INTERVAL
## 
## When schooling_c is OUTSIDE the interval [-51.10, -4.35], the slope of
## fossil_share_c is p < .05.
## 
## Note: The range of observed values of schooling_c is [-8.00, 4.88]
## 
## SIMPLE SLOPES ANALYSIS
## 
## When schooling_c = -3.155820e+00 (- 1 SD): 
## 
##                                Est.   S.E.   t val.      p
## ---------------------------- ------ ------ -------- ------
## Slope of fossil_share_c        0.01   0.00     3.04   0.00
## Conditional intercept          0.50   0.11     4.70   0.00
## 
## When schooling_c =  7.438494e-16 (Mean): 
## 
##                                Est.   S.E.   t val.      p
## ---------------------------- ------ ------ -------- ------
## Slope of fossil_share_c        0.01   0.00     6.78   0.00
## Conditional intercept          0.87   0.05    17.98   0.00
## 
## When schooling_c =  3.155820e+00 (+ 1 SD): 
## 
##                                Est.   S.E.   t val.      p
## ---------------------------- ------ ------ -------- ------
## Slope of fossil_share_c        0.01   0.00     6.41   0.00
## Conditional intercept          1.24   0.10    11.96   0.00
```

```r
cat("### Table 3a: Simple Slopes (4-digit precision)\n")
```

```
## ### Table 3a: Simple Slopes (4-digit precision)
```

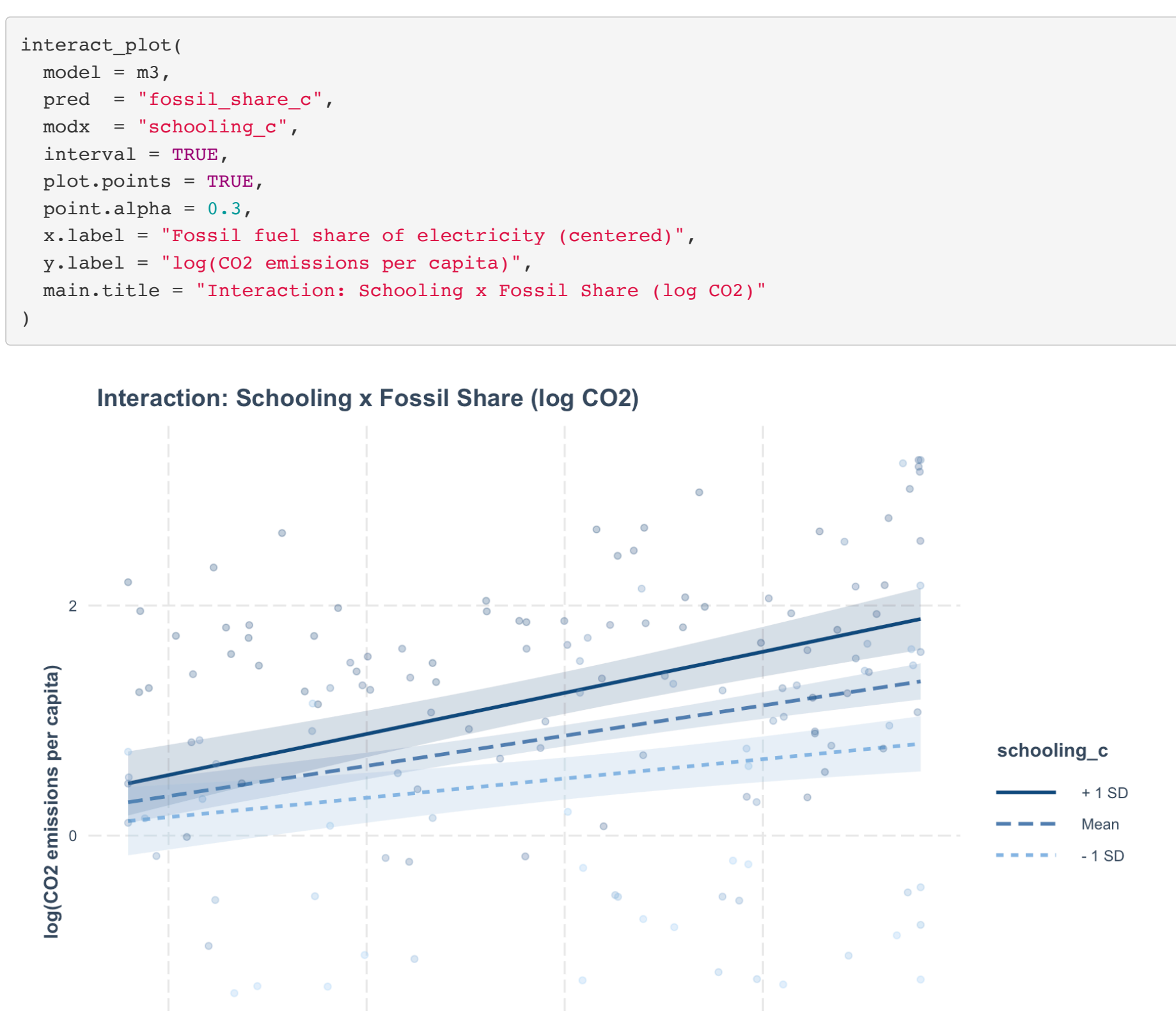
```r
s_slopes$slopes %>%
  as.data.frame() %>%
  kable(digits = 4, caption = "Slopes of fossil_share_c at different schooling levels")
```

Slopes of fossil_share_c at different schooling levels

Value of schooling_c	Est.	S.E.	2.5%	97.5%	t val.	p
-3.1558	0.0067	0.0022	0.0023	0.0111	3.0388	0.0028
0.0000	0.0105	0.0016	0.0074	0.0136	6.7754	0.0000
3.1558	0.0143	0.0022	0.0099	0.0187	6.4055	0.0000

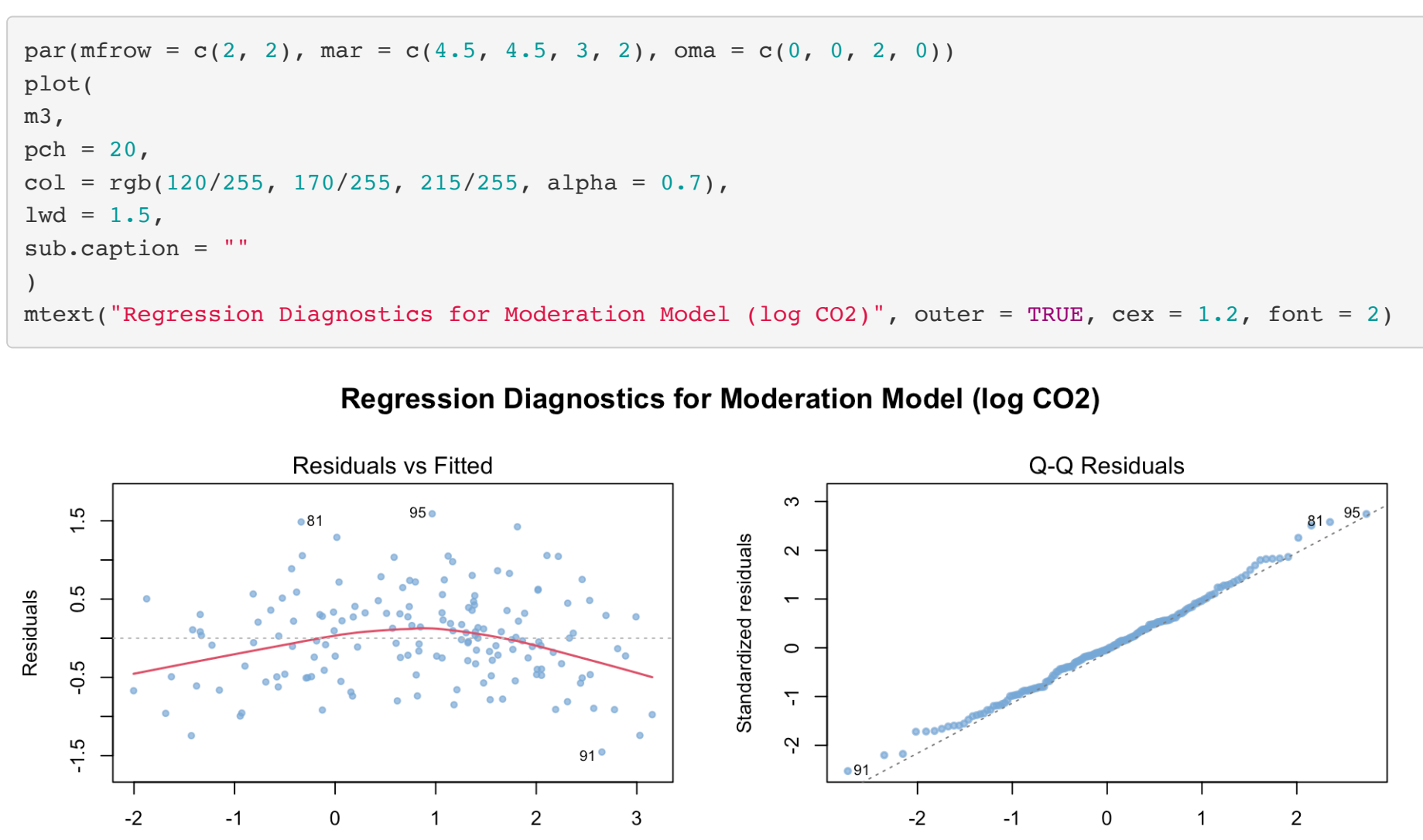
6.6 Interaction Plot

```r
interact_plot(
  model = m3,
  pred  = "fossil_share_c",
  modx  = "schooling_c",
  interval = TRUE,
  plot.points = TRUE,
  point.alpha = 0.3,
  x.label = "Fossil fuel share of electricity (centered)",
  y.label = "log(CO2 emissions per capita)",
  main.title = "Interaction: Schooling x Fossil Share (log CO2)"
)
```



7 Model diagnostics

```r
par(mfrow = c(2, 2), mar = c(4.5, 4.5, 3, 2), oma = c(0, 0, 2, 0))
plot(
m3,
pch = 20,
col = rgb(120/255, 170/255, 215/255, alpha = 0.7),
lwd = 1.5,
sub.caption = ""
)
mtext("Regression Diagnostics for Moderation Model (log CO2)", outer = TRUE, cex = 1.2, font = 2)
```



```r
par(mfrow = c(1, 1))
```

8 VIF and model comparison

```r
vif_m3 <- vif(m3)
vif_m2 <- vif(m2)
vif_comparison <- data.frame(
Variable = names(vif_m3),
VIF_with_Interaction = as.numeric(vif_m3),
VIF_without_Interaction = as.numeric(vif_m2[names(vif_m3)])
)

kable(vif_comparison, digits = 3, caption = "VIF Comparison (with vs without interaction)")
```

VIF Comparison (with vs without interaction)

Variable	VIF_with_Interaction	VIF_without_Interaction
fossil_share_c	1.014	1.012
ln_gdp	2.981	2.941
schooling_c	2.963	2.961
fossil_share_c:schooling_c	1.030	NA

9 F-test (Model 2 vs Model 3)

9 F-test (Model 2 vs Model 3)

```
f_test_m2_m3 <- anova(m2, m3)
kable(f_test_m2_m3, caption = "F-test: Compare Model 2 vs Model 3")
```

F-test: Compare Model 2 vs Model 3

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
156	55.25637	NA	NA	NA	NA
155	52.88496	1	2.371419	6.95037	0.0092328